Classical Field Theory — Exercise Sheet 9

Gauge identities and source conditions

Hints and answers

Use $A_{\mathrm{em}} = -\phi_{\mathrm{em}}\mathrm{d}t + \mathbf{A}\cdot\mathrm{d}x$, $\mathbf{E} = -\nabla\phi_{\mathrm{em}} - \partial_t\mathbf{A}$ and $\mathbf{B} = \operatorname{curl}\mathbf{A}$. Unless a weaker class is explicitly discussed, fields and coefficients are smooth. Compact support is stated separately in each problem.

Exercise 9.1 (The adjoint differentiates coefficients too). Let a smooth first-order density $L(x,\phi,\partial\phi)$ have Euler expressions E_a on an open coordinate region $U \subset \mathbb{R}^{1+d}$. Vary its action on a compact region $D \subset U$, with every parameter supported in the interior of D. For smooth parameters ϵ^α, consider the finite-order variation

$$\delta_\epsilon\phi^a = \sum_{|I|\le r} R^{aI}_\alpha(x, j^s\phi)\partial_I\epsilon^\alpha.$$

Here I is a derivative multi-index, each multi-index occurs once in the sum, and $j^s\phi$ denotes the field and all its derivatives through finite order s. The coefficients are smooth functions of these arguments. Assume the action is invariant under these variations for every compact parameter and every smooth field, without imposing the field equations.

(a) Derive the identity satisfied by the Euler expressions, including the variable coefficients. For the one-dimensional test operator $R\epsilon = (1+x^2)\epsilon_x + x\epsilon$, display R^*E explicitly. This test asks for its adjoint, not for a claim that every density has that symmetry.

(b) For Maxwell's transformation $\delta\phi_{\mathrm{em}} = -\epsilon_t$, $\delta\mathbf{A} = \nabla\epsilon$, derive the identity for the scalar and vector Euler expressions. With $\mathcal{M}_0 = \operatorname{div}\mathbf{E}$ and $\boldsymbol{\mathcal{M}} = \partial_t\mathbf{E} - \operatorname{curl}\mathbf{B}$, include a prescribed source $(\rho,\mathbf{J})$ and identify its necessary conservation condition. Explain the distributional interpretation for only C^2 potentials.

Hint. Apply the compact first-variation formula before moving derivatives off the parameter. Each integration by parts differentiates the full product $R^{aI}_\alpha E_a$.

Answer. $\sum_{a,I}(-1)^{|I|}\partial_I(R^{aI}_\alpha E_a) = 0$. The test operator gives $R^*E = -(1+x^2)E_x - xE$. For Maxwell, $\partial_t E_\phi - \operatorname{div}E_A = -(\rho_t + \operatorname{div}\mathbf{J})$, so invariance requires a conserved source.

Exercise 9.2 (Polarization and magnetization). Let $P, M \in C^\infty_c(\mathbb{R}^{1+3};\mathbb{R}^3)$ and define

$$\rho = -\operatorname{div}P, \qquad \mathbf{J} = \partial_t P + \operatorname{curl}M.$$

Prove conservation, zero total charge on each slice and $\int y\rho(t,y)\mathrm{d}^3y = \int P(t,y)\mathrm{d}^3y$. For any smooth potential, rewrite the integrated source coupling $\int(-\rho\phi_{\mathrm{em}} + \mathbf{J}\cdot\mathbf{A})\mathrm{vol}$ in terms of P, M and the fields, displaying why all discarded terms vanish. Does this construction assert that the potential of this source is unique?

Hint. Use the divergence of a curl, then coordinate integration by parts in space and time. The curl term transfers to $\mathbf{A}$ with a plus sign.

Answer. The source is conserved and neutral, its dipole is $\int P$, and its coupling is $\int(P\cdot\mathbf{E} + M\cdot\mathbf{B})\mathrm{vol}$. Gauge-related potentials are still not unique.

Exercise 9.3 (Vacuum duality with the Lorentzian star). For a vacuum Maxwell field, put $F' = \star F_{\mathrm{em}}$, using exactly the Hodge table of Lecture 9 and $\star^2 = -1$ on two-forms. Find the transformed electric and magnetic vectors, and verify all four vacuum component equations. Determine the effect of two and four successive transformations. Check the energy and Poynting densities for the constant vacuum fields $\mathbf{E} = (1, 0, 2)$, $\mathbf{B} = (0, 3, 0)$ before and after the map. These are densities, not finite total energies on $\mathbb{R}^3$. Does the same transformation preserve Maxwell's equations with a prescribed nonzero electric current three-form $\mathcal{J}$ and no magnetic source?

Hint. Express the starred two-form again in the basis $\mathrm{d}x^i \wedge \mathrm{d}t$ and the spatial magnetic wedges. Their order matters.

Answer. $(\mathbf{E}, \mathbf{B}) \mapsto (-\mathbf{B}, \mathbf{E})$; twice gives $(-\mathbf{E}, -\mathbf{B})$ and four times the identity. Both examples have energy density 7 and flux $(-6, 0, 3)$. For nonzero electric source, $\mathrm{d}F' = \mathcal{J}$ rather than zero.

Exercise 9.4 (Conservation does not remove a time-face term). Let $D = [t_0, t_1] \times \mathbb{T}^3$, $t_1 > t_0$, and use the Maxwell density $L = \frac{1}{2}(|\mathbf{E}|^2 - |\mathbf{B}|^2) - \rho\phi_{\mathrm{em}} + \mathbf{J} \cdot \mathbf{A}$. Take smooth periodic fields and a prescribed smooth conserved source. Allow a smooth periodic gauge function $\chi(t, x)$ with nonzero time-face values.

(a) Find the exact action change, retaining the time faces and explaining the cancellation on opposite spatial faces.

(b) Set $\rho = \sin x^1$, $\mathbf{J} = 0$, $\phi_{\mathrm{em}} = \sin x^1$, $\mathbf{A} = 0$, and $\chi = \gamma t \sin x^1$, where real $\gamma \neq 0$. Verify Maxwell's equations before and after the transformation and compute the nonzero action change. Why is this consistent with Lecture 9's compact-parameter gauge invariance?

Hint. The kinetic density is unchanged. Rewrite $\rho\chi_t + \mathbf{J} \cdot \nabla\chi$ as a spacetime divergence minus $\chi(\rho_t + \operatorname{div} \mathbf{J})$.

Answer. $S[A_{\mathrm{em}} + \mathrm{d}\chi] - S[A_{\mathrm{em}}] = [\int_{\mathbb{T}^3} \rho\chi \mathrm{d}^3x]_{t_0}^{t_1}$. The example has unchanged fields $\mathbf{E} = (-\cos x^1, 0, 0)$, $\mathbf{B} = 0$ and action change $4\pi^3\gamma(t_1 - t_0)$.